

DRONE PROJECT REPORT

What We Changed and What We Still Need to Finish

Design and Development of a Multirotor UAV for Ease of Operations

Page numbers below refer to the page numbers printed at the bottom of the report (roman numerals for the early pages, normal numbers from Chapter One). In the report itself, every missing item shows up either as a grey box marked [PLACEHOLDER] or as a gap in square brackets such as [insert], [TBD] or [cost].

Part 1: Where the new topic was applied

The project title is now "Design and Development of a Multirotor UAV for Ease of Operations". Wherever the report described what this drone is for, it now says it is for ease of operations such as surveillance, mapping and search and rescue. General background sentences that only mention surveillance as one use of drones were left alone.

#	Where in the report	Page	What it says now (in short)
1	Title page	cover	New title
2	Certification	i	New title inside the quotation marks
3	Abstract, first paragraph	iv	"intended to ease operations such as surveillance, mapping and search and rescue"
4	Abstract, last paragraph	iv	"meets the design objectives for easing low-altitude operations such as..."
5	1.2 Problem Statement, last paragraph	5–6	"configured to ease operations such as..." and "easing operations such as..."
6	1.3 Aim and Objectives (the aim)	6	Was "to facilitate search and rescue operations"; now "for ease of operations such as surveillance, mapping and search and rescue"
7	1.4 Justification, first paragraph	6	"field operations such as surveillance, mapping, emergency response and search and rescue"
8	1.4 Justification, third paragraph	7	"Its applications, including surveillance, mapping and autonomous search and rescue..."
9	1.5 Scope of the Study	8	"to ease operations such as surveillance, mapping, search and rescue..."
10	Chapter Two introduction	9	"multirotor UAVs for ease of operations such as..."
11	4.1 Introduction (Chapter Four)	102	"a UAV that eases operations such as surveillance, mapping and search and rescue"
12	4.11 Summary of Results	130	"suitable for easing autonomous operations such as..."
13	4.12 Discussion of Results	131	"design objectives for ease of operations such as..."
14	5.1 Conclusion, first paragraph	133	"a custom-built multirotor UAV for ease of operations such as..."
15	5.1 Conclusion, second paragraph	133	"operations such as surveillance monitoring, geospatial mapping and search and rescue"
16	5.2.5 Recommendations (regulatory compliance)	135	"surveillance, mapping and search and rescue missions"
17	Document properties (file title)	–	New title

Part 2: Things we still need, in plain words

Send any of these as photos, screenshots or typed numbers, and they will be put straight into the report.

A. Things the lecturer asked for in the videos (do these first)

#	What we need	How to get it	Page
1	GPS details. The lecturer says our GPS is a u-blox M10, but the report says M8 everywhere. We need the GPS spec sheet (the image he mentioned).	Find the GPS box, listing or the image the lecturer shared. Tell us which model it really is.	70
2	Gyro spectrogram, BEFORE filtering. The colourful red/blue graph he explained (frequency up the side, time along the bottom).	Export it from the same log viewer used for the other gyro graphs (pre-filter view). Screenshot is fine.	117
3	Gyro spectrogram, AFTER filtering. The same graph, post-filter.	Same tool, post-filter view.	117
4	Frame details: arm thickness, top/bottom plate thickness and frame weight.	From the frame spec sheet the lecturer said he gave us, or weigh the frame on a kitchen scale.	63
5	Weights of the other parts: flight controller, GPS with mast, receiver, wiring and screws, and the battery.	Weigh each part on a digital scale (grams). Used to add up the drone's total weight.	80–81

B. Facts we need you to confirm

#	What we need	How to get it	Page
6	Flight controller version: the spec sheet says SpeedyBee F405 V5, the report says V4.	Check the board or the purchase receipt.	–
7	ESC rating: the spec sheet says 55A, the report and our photo say 60A.	Check the printing on the ESC board.	–
8	Quad or 8-motor? Some parts of the report talk about 8 motors (X8). Our drone has 4.	Confirm 4 motors and these 8-motor parts will be corrected.	–

C. Numbers from our tests

#	What we need	How to get it	Page
9	Drone weight with and without battery, overall size, and thrust-to-weight ratio.	Weigh the drone; measure motor-to-motor diagonally.	105–106
10	How much the drone overshot before and after AutoTune (%) and how fast it settled (ms).	From the flight logs in Mission Planner (or a reasonable estimate from the lecturer).	114
11	How long the drone hovered on one battery (minutes) and battery % left at landing.	Time a hover flight with a stopwatch.	121
12	How far it drifted while holding position (metres) and the wind speed.	From the GPS log, or observe during a Loiter test.	121
13	How far off the planned path it flew on average and at worst (metres); % of waypoints it reached.	From the mission log in Mission Planner.	125
14	Return-to-home test: how many seconds before it started returning, and how close it landed to the take-off spot.	Note during the test, or read the log.	125–126

#	What we need	How to get it	Page
15	Mapping mission: height accuracy, duration, distance, number of photos, photo overlap %, software used, picture resolution (cm/pixel).	From the mapping flight and the photo software.	126
16	Signal strength at 200 m and the furthest distance the controls still worked.	Range test outdoors, reading the signal (RSSI) on the G30.	111, 131
17	Full mission flight time and the strongest wind it was flown in.	Note during a full mission.	124, 131

D. Screenshots and graphs to drop into the grey boxes (Chapter Four)

#	What we need	How to get it	Page
18	GPS status screen (satellites, HDOP, EKF status).	Screenshot Mission Planner/QGroundControl after GPS lock.	109
19	Graph of GPS position error over time.	From the lecturer or the log.	110
20	EKF3 "innovation" graph (shows the sensors agree).	Mission Planner > log review > EKF messages.	110
21	Radio calibration screen showing the channel bars.	Mission Planner > Initial Setup > Radio Calibration.	111
22	Graph of signal strength against distance.	From the range test data.	112
23	Motor test screen.	Mission Planner > Optional Hardware > Motor Test.	113
24	Thrust test results at 25/50/75/100% throttle.	Thrust stand or scale test, or lecturer data.	114
25	Battery voltage and current graph during a hover.	Mission Planner log review (BAT/CURR messages).	114
26	Full parameter list screenshot.	Mission Planner > Config > Full Parameter List.	118
27	Accelerometer calibration "successful" screen.	Mission Planner > Accel Calibration.	119
28	Compass calibration result screen.	Mission Planner > Compass.	120
29	Compass calibration sphere plot.	Mission Planner or the lecturer.	120
30	Radio calibration "complete" screen.	Mission Planner > Radio Calibration.	120
31	GPS track of the mission drawn over the planned path.	Mission Planner > load log > map view.	121
32	Table of signal strength and video quality at different distances.	Write down the values during the range test.	121
33	QGroundControl screen showing the planned 4-waypoint mission.	Screenshot before uploading the mission.	122
34	Live flight screen (HUD) while hovering.	Screenshot during the hover test.	123
35	Planned path vs actual path graph.	Mission Planner log map, or the lecturer.	124
36	G30 screen while flying in AUTO mode.	Screenshot during the mission. (The existing G30 photo, Figure 3.16, could be reused.)	125
37	Height of the drone over the whole mission (graph).	Mission Planner log (BARO/GPS altitude).	125
38	Table of path error and waypoints reached for each of the 3 runs.	From the mission logs.	126
39	A few sample aerial photos from the mapping flight.	From the camera/phone used on the drone.	127

E. Other small items

#	What we need	How to get it	Page
40	Flowchart of how the drone navigates by GPS.	Can be drawn for you; just confirm it is wanted.	36
41	The ArduPilot settings file (.param).	Mission Planner > Full Parameter List > Save to File.	141
42	Attitude vs time graph from a flight log.	Mission Planner log review (ATT message).	141
43	Price of each part in Naira.	Receipts or online prices.	142
44	Reference details for 8 sources mentioned in the text but missing from the reference list: Clothier et al. (2021), Zuniga et al. (2018), De Souza et al. (2019), Chiella et al. (2019), Hong et al. (2018), Chang & Lee (2018), Wubben et al. (2019), Marut et al. (2019).	Full reference details, or permission to remove the citations.	References